SURJECTIVITY ON OBJECTS IS STABLE UNDER BASE CHANGE

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Editorial note. No omission markers in-span; proof complete as written.

Lemma 0.1. *Let $p : \mathcal{X} \to \mathcal{Y}$ and $q : \mathcal{Z} \to \mathcal{Y}$ be 1-morphisms of categories fibred in groupoids over $(Sch/S)_{fppf}$. If $p : \mathcal{X} \to \mathcal{Y}$ is surjective on objects, then so is the base change $p' : \mathcal{X} \times_{\mathcal{Y}} \mathcal{Z} \to \mathcal{Z}$ of p by q.*

Proof. This is formal. Let z be an object of $\mathcal{Z}$ over a field k. As p is surjective on objects there exists an extension K/k and an object x of $\mathcal{X}$ over K and an isomorphism $\alpha : p(x) \to q(z)|_{\mathrm{Spec}(K)}$. Then $w = (\mathrm{Spec}(K), x, z|_{\mathrm{Spec}(K)}, \alpha)$ is an object of $\mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$ over K with $p'(w) = z|_{\mathrm{Spec}(K)}$. □

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